

Design Thinking in Practice: Lessons from Context-Sensitive Implementation in SMEs

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Abstract

This study explores how Design Thinking is implemented in practice within small and medium-sized enterprises (SMEs), and which contextual factors shape its success. It addresses the research question of how Design Thinking is implemented in SMEs and what contextual factors influence its outcomes. Using semi-structured expert interviews with seven Design Thinking consultants, the study employs a qualitative, exploratory design and applies the Gioia methodology for inductive analysis. The analysis identifies six thematic dimensions that explain how Design Thinking becomes actionable in SMEs, ranging from strategic alignment to long-term embedding. The findings show that effective SME adoption of Design Thinking hinges on leadership commitment, early managerial involvement, and cross-functional collaboration, and is characterized by pragmatic adaptations such as shorter workshop formats distributed over several weeks, low-cost prototyping (sketches, mock-ups, simple digital artefacts), and an iterative, context-sensitive integration into everyday routines rather than adherence to linear stage models. Sustainability (in terms of organizational capacity and resilience) frequently emerges as a secondary outcome rather than a predefined objective. These practices enable SMEs to manage operational constraints while achieving user-centric problem framing and faster learning loops. The study contributes to the growing body of research on innovation management in SMEs by providing an empirically grounded account of SME-specific Design Thinking adaptations and offering actionable implications for managers, consultants, and policymakers.

Keywords: design thinking, SME, innovation management, qualitative research, innovation capabilities.

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1 Introduction

Small and medium-sized enterprises (SMEs¹) are a large part of the global economy. They are responsible for job creation and contribute significantly to economic growth. They have to struggle to remain competitive, and current turbulent and dynamic markets expose them to increasing risks (Gherghina, Botezato, Hosszu, & Simionescu, 2020). SMEs tend to have, however, access to less resources and therefore struggle to implement planned organizational transformation. The

1. In this study, SMEs are defined based on the European Commission criteria: companies with fewer than 250 employees.

adoption of innovation-oriented practices, remains relatively low in SME contexts due to their limited awareness, expertise, and resources (Taneja, Pryor, & Hayek, 2016).

Design Thinking, understood as a user-centered, iterative, and creative problem-solving methodology that may help promoting innovation, has been used in different industries, in product development, service design, and organizational change (Jaskyte & Liedtka, 2022). While large firms frequently have the capacity to integrate Design Thinking, SMEs tend to operate with flatter structures, and a greater reliance on intuitive decision-making (Hookway et.al., 2019).

Recent studies have noted that the adoption of Design Thinking in SMEs requires methodological adjustments, as well as cultural and organizational transformation, including alignment with innovation strategy and stakeholders' goals. In some cases, it focuses on sustainability, hereby understood as the long-term capacity of the company to innovate and adapt (Amuy, Jabbour, Jabbour, & Kannan, 2017; Adams, Jeanrenaud, Bessant, Denyer, & Overy, 2016). Nonetheless, academic frameworks to support SMEs in putting Design Thinking into practice are scarce (Ghag, Acharya, & Khanapuri, 2022), and many companies struggle to include design-led thinking in their routines (Carlgren, Rauth, & Elmquist, 2016).

To fill in this gap, this study explores how Design Thinking is implemented in practice within SMEs.

Our key research question is: How is Design Thinking implemented in SMEs, and what contextual factors influence its success? We collect and analyze data on the perspectives of expert consultants with experience supporting innovation in different SME contexts and we identify key enablers and barriers to its adoption. Results show how Design Thinking is interpreted and adapted in SME contexts to promote innovation and build competitive advantage.

2 Literature Review

2.1 Design Thinking as a Driver of Innovation in SMEs

Design Thinking has been studied, in academic and professional contexts, as a human-centered, creative approach to problem-solving. The literature distinguishes between two main discourses: the *design discourse*, which focuses on theoretical development, and the *management discourse*, which emphasizes the application of Design Thinking in business contexts (Carlgren, Elmquist, & Rauth, 2016; Carlgren et al., 2016; Johansson-Skoldberg, Woodilla, & Cetinkaya, 2013).

While much of the research has focused on large organisations that possess the complexity and resources to implement such methodologies (Kotler & Keller, 2016), the context of SMEs presents distinct challenges. SMEs often have limited financial, human, and knowledge resources, which may endanger the adoption of innovation practices like Design Thinking (Demir, Saur-Amaral, Polónia, 2023a; Fischer et al., 2020). The practical advantages that larger firms leverage, e.g., extensive design teams, structured innovation labs, or complex customer segmentation, are rarely available to SMEs.

Nevertheless, Design Thinking is not irrelevant to SMEs. On the contrary, several authors argue that due to their flatter hierarchies, agility, and closer customer relationships, SMEs can benefit from user-centered innovation approaches (Schweitzer, Groeger, & Sobel, 2016; Roberts et. al., 2016; González-Calleros et al., 2024). However, measuring the benefits or aligning Design Thinking with existing routines and structures is still a challenge. Barriers include limited expertise, scepticism among leadership, and uncertainty regarding Design Thinking's value in lean environments (Beckman, 2020; Carlgren et al., 2016; Kabir & Mukerjee, 2024; You, 2022).

The implementation of Design Thinking in SMEs has also received comparatively limited academic attention (Gasda & Fueglistaller, 2016). Kammerlander et al. (2015) attribute this

to a general underestimation of SMEs' innovation potential. Moreover, many SMEs prioritise operational continuity over strategic innovation, often delaying the adoption of structured methods such as Design Thinking (Dell'Era et al., 2020). Nonetheless, studies show that SMEs are open to experimentation and learning, provided the methods are adapted to their specific needs (Shapira, Ketchie, & Nehe, 2017; Tuffuor et al., 2025).

To support understanding of how Design Thinking may be operationalised in SMEs, Table 1 presents a synthesis of strategic integration pathways identified in the literature. These show how Design Thinking can align with innovation management objectives and support broader cultural and strategic shifts within SME environments. The table also illustrates how SMEs can advance from informal innovation initiatives to more structured, design-led practices without the infrastructure of large organisations.

Table 1. Strategic Pathways for Integrating Design Thinking in SMEs.

Strategic Focus	Organisational Practices	Design Thinking Enablers	Expected Outcomes
Embedding Innovation in Organisational Strategy	Integrate Design Thinking into strategic planning and corporate culture	Align Design Thinking with strategic goals; embed in decision-making	Enhanced innovation capacity and sustained innovation culture
Promoting a User-Centred Mindset	Allocate resources and create design-oriented teams	Emphasise empathy, user needs, and collaboration	Improved customer alignment and competitive differentiation
Leveraging Design Thinking as a Growth Catalyst	Apply Design Thinking to complex challenges; develop employee skills	Recognise Design Thinking as strategic; integrate into training	Business growth, adaptability, innovation in uncertain contexts
Opening Innovation to External Collaboration	Engage clients, suppliers, and institutions; simplify processes	Foster cross-boundary collaboration and co-creation	Accelerated innovation cycles and responsiveness

Source: Adapted from Demir and colleagues (2023b)

2.2 Strategic and Organisational Integration of Design Thinking in SMEs

In recent years, scholars have emphasized the need to move beyond viewing Design Thinking as a toolkit, proposing instead that it be adopted as a mindset and strategic orientation (Liedtka & Ogilvie, 2011; Olivares Ugarte & Bengtsson, 2024). When integrated into strategy formulation and execution, Design Thinking contributes to problem-framing, idea generation, and alignment with organisational vision (Liedtke, 2018; González-Calleros et al., 2024). This is in line with authors who argue that Design Thinking is useful to address "wicked problems", i.e., complex, multi-dimensional challenges that resist linear solutions (Booth et.al., 2021).

Integrating Design Thinking into SME innovation management involves aligning it with existing goals and processes, dedicating spaces and resources for experimentation, and recognising Design Thinking as a strategic capability rather than an operational tool (Dell'Era et al., 2020; Wrigley, Nusem, & Straker, 2020). For this to happen in SMEs, adaptation is crucial: methods need to be scaled, simplified, and embedded in routines, often with support from external expertise.

The literature emphasizes the value of external knowledge networks as strategic mechanisms for SMEs to deal with internal resource constraints. Collaborations with universities, consultants, and customers enable SMEs to access complementary expertise and capabilities (Gao & Yu, 2023; Vătămănescu et al., 2020). Strategic collaboration and knowledge sharing can enable SMEs to access tools, skills, and perspectives that would otherwise be unavailable to them.

Table 2 illustrates how Design Thinking can be strategically integrated into SME processes, showing its potential to support differentiation, improve customer experience, and increase internal efficiency. The table proposed roles for Design Thinking beyond isolated innovation projects, showing its relevance to broader business development.

Table 2. Roles of Design Thinking in Addressing Key Business Needs in SMEs.

Business Focus Area	Strategic Objectives	Design Thinking Applications	Expected Outcomes
Business Challenges	Efficient Innovation	Incorporate design contributions into strategic planning; explore novel business models	Innovative offerings, improved market positioning, customer-centric solutions
Strategic / Business Aim	Design Strategy Planning	Define and articulate brand identity; integrate design in strategy	Stronger brand equity and alignment with user expectations
Offerings	Practical Design	Apply iterative prototyping and validation with customers and stakeholders	Validated, user-relevant solutions with higher adoption rates
Revenues	Leveraging Design for Competitive Advantage	Translate research into premium features; use emotional and cultural contributions	Increased willingness to pay, stronger customer loyalty
Margins & Savings	Simplification through Design	Implement modular design; reuse design assets across offerings	Cost efficiency, scalability, and consistent user experience

Source: Adapted from Sato (2009), restructured and expanded by the authors.

In summary, while the literature supports the strategic value of Design Thinking in SMEs, it also points to a lack of operational frameworks and guidance for smaller firms. This reinforces the need for empirical studies that explore how SMEs implement Design Thinking under real constraints, which is the focus of this study.

3 Methodology

3.1 Research Design

The goal of this study is to explore how Design Thinking is implemented in real-world SME contexts, by examining the experiences, practices, and perspectives of professionals with direct involvement in such processes. Considering the exploratory nature of the research and the goal of understanding context-specific mechanisms, a qualitative design was adopted (Brinkmann & Kvale, 2015; Flick, 2022). The intention was not to test predefined hypotheses, but to collect and analyse data from expert practice with SMEs.

Accordingly, we used semi-structured interviews with experts as our primary data source. This approach is suited to research complex, contemporary phenomena and allows for the identification of recurring patterns across different perspectives, supporting analytical generalization (Yin, 2018).

3.2 Sampling and Expert Selection

Participants were selected through purposive sampling (Robinson, 2014) targeting professionals with proven experience in implementing Design Thinking in SMEs. Inclusion criteria included seniority, consultancy experience, and demonstrated involvement with SME innovation processes.

Each expert brings extensive knowledge of design-led innovation and direct experience with the challenges and constraints faced by SMEs. The sampling strategy focused on their cumulative practice-based knowledge across multiple SME contexts rather than on single-firm case analysis.

The final sample consisted of seven experts, all of whom have guided SME innovation projects. A summary of their profiles is presented in Table 3, including role, experience, and area of expertise.

Table 3. Roles of Design Thinking in Addressing Key Business Needs in SMEs.

Interviewee	Gender	Designation	Role in the company	Years of experience	Started Design Thinking	Description
A	Male	CEO & Owner	Managing Director	8	2017	Experienced Design Thinking consultant with a creative mindset and ability to develop innovative approaches for complex problem-solving.
B	Female	CEO & Owner	Consultant & Design Thinking Expert	14	2010	Consultant with extensive experience in applying Design Thinking to generate effective, practical solutions for complex problems.
C	Female	CEO & Owner	Consultant & Design Thinking Expert	13	2012	Expert known for translating complex challenges into innovative outcomes using Design Thinking.
D	Male	CEO & Owner	Management Consultant	13	2014	Expert recognized for his innovation expertise and successful implementation of new ideas and proven track record in developing innovative solutions.

Interviewee	Gender	Designation	Role in the company	Years of experience	Started Design Thinking	Description
E	Male	CEO & Owner	Consultant & Design Thinking Expert	15	2006	Specialist in aesthetic and functional design with broad Design Thinking experience, known for creative development and implementation.
F	Male	Executive Manager	Design Director	7	2015	Expert with extensive experience in applying Design Thinking, Design Sprints and SME innovation.
G	Female	Executive Manager	Management Consultant	10	2013	Creative consultant with extensive knowledge to applying Design Thinking to deliver innovative, user-focused solutions.

Source: Developed by Author

This expert-centric sampling ensures that the results obtained are based on hands-on experience with Design Thinking implementation in SMEs across diverse sectors.

3.3 Data Collection

Data were collected through semi-structured expert interviews. An interview guide was developed and refined through pilot testing prior to data collection. It included open-ended questions, prompts, and exploratory scenarios related to Design Thinking implementation. A combination of visual and interactive techniques was employed, such as coloured cards, MIRO boards, and scenario walkthroughs. Each interview lasted approximately one hour. Interviews were conducted online and in person, depending on experts' availability. Audio recordings were made with participant consent, and all interviews were transcribed verbatim.

3.4 Data Analysis

The interview transcripts were analyzed using the Gioia methodology (Gioia, Corley, & Hamilton, 2013; Magnani & Gioia, 2023), a structured, widely adopted approach to inductive qualitative research. This method involves three levels of analysis: first-order concepts (informant terms), second-order themes (researcher interpretation), and aggregate dimensions.

Coding was performed using NVivo software and followed a constant comparison logic to identify patterns within and across interviews. This iterative process ensured that emerging categories were empirically grounded. Analytical rigour was enhanced through iterative peer review and feedback from selected participants. This iterative triangulation strengthened the credibility and robustness of the resulting data structure.

While the sample was small, as is common in qualitative research, experts' interviews allowed for the identification of practical mechanisms of Design Thinking implementation in SMEs, as experienced by those with repeated and varied exposure to such contexts.

4 Results and Findings

The findings of this study illustrate how Design Thinking is understood and implemented in SMEs, based on the perspectives of expert consultants. Applying the Gioia methodology enabled the identification of recurring themes and patterns across the interviews. These are organised into aggregate dimensions derived from first-order codes and second-order interpretative categories.

A consistent view among the interviewees was that Design Thinking is both a mindset and a structured approach. Experts described it as a means of breaking down complex problems and developing innovative, user-centered solutions. As participant C noted:

"Design Thinking is first and foremost a mindset. It's the way people think and act... it helps enable innovations more quickly."

This mindset encompasses empathy, collaboration, and experimentation, all core principles consistently highlighted in participant accounts. Design Thinking was also described as adaptable and context-sensitive, particularly relevant for SMEs operating with limited resources and requiring pragmatic innovation approaches.

Participants repeatedly emphasised the need to translate Design Thinking into daily SME routines. Unlike large companies with dedicated innovation teams, SMEs rely on multifunctional staff, requiring flexibility in how Design Thinking is introduced and applied.

"Design Thinking is a process or a method with which you can break down complex issues into feasible ideas..." (Participant A)

"It's a toolbox... that enable me to innovate better, generate ideas, to frame problems." (Participant E)

4.1 Expert Perspectives on Success Factors

To assess alignment between literature and practice, experts were asked to comment on success factors identified in prior studies. Their responses are synthesised in Figure 1, which visualises the relevance of these factors as assessed by experts².

The data analysis resulted in a set of interrelated dimensions that illustrate how Design Thinking is applied in SMEs. These include elements related to vision, where Design Thinking is seen as a guiding philosophy embedded in organisational purpose and strategic direction; mission, which reflects the broader goals of empathy, communication, and long-term orientation in Design Thinking adoption; and strategy, where Design Thinking is integrated into innovation practices, user research, and operational methods. Across these layers, experts emphasised the importance of aligning Design Thinking with the company's values and decision-making structures. Rather than following a rigid methodology, SMEs adapt Design Thinking principles through iterative learning, lightweight experimentation, and stakeholder engagement. The unified dimension that emerges captures the practical, evolving, and integrated nature of Design Thinking in these contexts.

2. "Most important" refers to factors identified by experts as essential; "Practical experience" indicates factors frequently observed in SME practice; and "Not relevant" covers factors considered of minor importance.

These themes are presented in Table 4, which presents the structured data derived from the thematic coding process and illustrates the progression from first-order codes to second-order themes and aggregate dimensions, based on the Gioia methodology.

Table 4. Application of Design Thinking and Innovation in SMEs.

First-order Codes	Second-order Codes	Summarized Dimension
Vision	Design Thinking as a visionary approach Aiming for global improvement Evolved into our guiding philosophy A contemporary and relevant vision Fostering a mindset shift Incorporated into corporate frameworks	Application of Design Thinking and Innovation in SMEs
Mission	Advancing global well-being Cultivating empathy and effective communication within organizations Aligning with current missions and objectives Embracing Design Thinking as a holistic mindset Integrating the mindset throughout corporate structures Enhancing understanding and application at a profound level Employing synthesis to identify problem areas Utilizing visualization techniques Incorporating financial modeling into the process Empowering clients to develop necessary skills Providing an end-to-end experiential journey	
Strategy	Enhance empathy and communication within companies Adopt Design Thinking as a core philosophy Develop innovative product features with multiple applications Engage customer representatives Establish dedicated Design Thinking structures and labs Conduct intensive workshops to cultivate the mindset Leverage Agile, Scrum, or Design Sprints Incorporate financial modeling for decision-making Utilize diverse research methods (competitive research, surveys, interviews) Deliver an end-to-end user-centric experience	

Source: Developed by Author

In addition, Appendix 1 provides a focused view of how a sustainable Design Thinking approach can be implemented across different SME contexts. Based on expert contributions, it explores the purpose, process, and long-term use of Design Thinking in practice. This additional data reinforces the idea that Design Thinking adoption in SMEs goes beyond technical application, often reflecting deeper organisational ambitions and cultural shifts.

4.2 Practical Success Factors and Divergences from Theory

The interviews also identified several SME-specific challenges and adaptations that extend beyond the theoretical success factors typically found in the literature. Experts noted that although leadership involvement is widely recognised as important, managers in SMEs are often only marginally involved at the early stages of Design Thinking initiatives. This limits strategic alignment and reduces internal commitment. As one expert explained:

“If management is not in from the beginning, Design Thinking remains a side activity and is not taken seriously in the company” (C).

Another recurring theme was the perception of Design Thinking as a superficial or creative-only exercise, especially among SME leaders unfamiliar with the methodology. This perception often required consultants to reframe Design Thinking as a rigorous and structured problem-solving approach:

“Many people think Design Thinking is just about sticky notes and idea storms. We need to show them it’s a disciplined process” (F).

In response to time and resource constraints, SMEs often adapted the Design Thinking process to fit their context. For example, workshops were split into shorter sessions of two to three hours across multiple weeks to avoid disrupting operations. Similarly, prototyping was typically low-cost and rapid, relying on sketches or simple digital mock-ups rather than elaborate models. As one expert explained:

“We rarely run full-day sessions. Instead, we do two or three hours per week over a month, because people cannot leave operations for longer” (B).

These findings illustrate the importance of contextualising Design Thinking for SME realities. Enablers unique to SMEs include close relationships with customers, which facilitate real-time feedback, and flatter hierarchies, which speed up internal decision-making. However, these same characteristics can lead to role overload, with employees contributing to innovation efforts while managing their regular tasks:

“In SMEs, people wear many hats – they can contribute to Design Thinking, but it also means they are stretched thin” (D).

Taken together, the findings illustrate how SMEs appropriate Design Thinking through an adaptive, pragmatic lens. While the literature often frames Design Thinking as a formal and linear methodology, its application in SMEs is more fluid, shaped by constraints and opportunities specific to small firm contexts. As one expert observed:

“Without leadership pushing it, Design Thinking never becomes more than a one-off workshop – but if strategy and culture support it, it can really change how people work” (E).

Figures 1 and 2 illustrate the experts’ evaluations of success factors and how these are grouped into practical clusters.

Figure 1: Perceived relevance of success factors, based on expert evaluation

Source: Developed by Author



Figure 1. Perceived relevance of success factors, based on expert evaluation
Source: Developed by Author

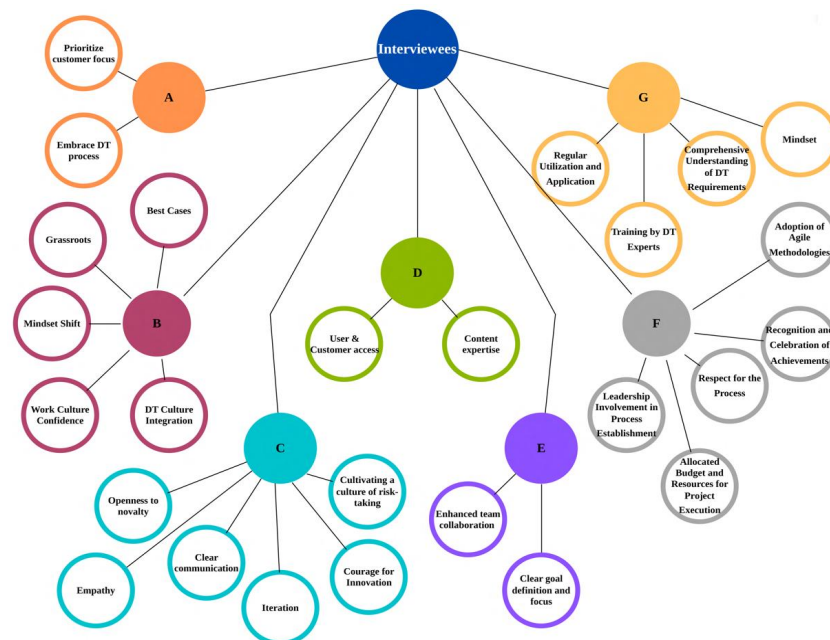


Figure 2. Experts Success Factors derived from practical experiences

Source: Developed by Author

Note: Labels A–G represent anonymized interviewees corresponding to Table 3. The thematic clusters are described in the accompanying text and in Table 4.

Figure 1 presents the perceived relevance of various success factors from expert interviews. It distinguishes between those considered essential, those frequently observed in practice, and those viewed as less relevant in the SME context. This visual helps clarify how practitioners prioritise and interpret the conditions for successful Design Thinking adoption.

Figure 2 builds on this by grouping the identified factors into thematic clusters derived from the interview analysis. These clusters include strategic orientation, leadership commitment, organisational culture, collaborative dynamics, and resource constraints. The figure serves as a visual synthesis of how these elements interact in practice, offering a grounded representation of success conditions for Design Thinking implementation in SMEs. Rather than serving as a formal conceptual model, these figures reflect a practitioner-informed overview of what drives (or hinders) effective Design Thinking integration. They reinforce the study's main conclusion: that Design Thinking implementation in SMEs is shaped by a combination of enabling and constraining factors, requiring flexible, context-specific adaptation.

5 Discussion

This study provides a broad understanding of how Design Thinking is implemented in SMEs and which contextual factors shape its adoption. By integrating the literature and the analysis of expert interviews, the research identifies enabling conditions and the practical adaptations that allow Design Thinking to contribute to innovation outcomes in resource-constrained environments such as SMEs.

5.1 Scientific Contributions

This study contributes to the literature on design thinking and innovation management in SMEs in three main ways: theoretical, methodological, and practical.

First, the theoretical contribution lies in offering an empirical account of Design Thinking implementation in SMEs, a topic still underexplored in comparison to research on large firms or start-ups. Existing studies often describe Design Thinking as a linear and structured process institutionalised through dedicated teams and extensive resources (Carlgren, Elmquist, & Rauth, 2016). In contrast, this research demonstrates that SMEs interpret and apply Design Thinking in flexible and iterative ways that align with their operational routines, limited resources, and organisational culture.

It thereby extends prior research by showing that contextual conditions such as organisational size, leadership involvement, and internal culture act as key moderators of Design Thinking practice. These findings refine design-led innovation theory for smaller firms and reinforce the argument of Eisenhardt and Graebner (2007) that theory building should be grounded in empirical reality. The results also complement Sharma et al. (2022) by reaffirming the relevance of managerial commitment and internal culture as decisive enablers of innovation adoption in SMEs.

The methodological contribution results from the systematic application of the Gioia methodology to the study of innovation processes in SMEs, an approach that has been rarely applied in this domain. The three-level data structure consisting of first-order concepts, second-order themes, and aggregate dimensions ensures analytical transparency and theoretical rigour. This approach offers a replicable and transparent framework for future qualitative research on design-led innovation in SMEs.

The practical contribution of this study lies in translating the empirical insights into a framework of SME-specific Design Thinking adaptations. The findings illustrate how SMEs modify Design Thinking through shorter workshop formats, simplified prototyping, and incremental experimentation while maintaining methodological coherence. These results provide actionable knowledge for managers, consultants, and policy makers who seek to embed Design Thinking in resource-constrained environments and promote innovation through design-led processes.

Although sustainability is not always a predefined goal in SMEs, the findings demonstrate that Design Thinking can promote more responsible and future-oriented solutions when applied consistently. Sustainability therefore emerges as an indirect but valuable outcome of design-led innovation, linking organisational learning with broader strategic and societal objectives.

In conclusion, this study advances theoretical understanding by contextualising Design Thinking for SMEs, contributes methodologically by employing a rigorous inductive framework, and provides practice-oriented guidance for fostering innovation in smaller organisations operating under real-world constraints.

5.2 Practical Implications

From a managerial perspective, the study offers practical guidance for SME managers, innovation consultants, and policymakers. The findings show that Design Thinking can be integrated into SMEs without requiring extensive resources, provided that the approach is tailored to the firm's needs and introduced incrementally.

For SME managers, key recommendations include starting with small, manageable Design Thinking initiatives, ensuring leadership involvement from the outset, and integrating Design Thinking into day-to-day routines. Simplified prototyping, short workshop formats, and low-cost tools allow Design Thinking to be implemented without major disruption to ongoing operations. Managers play a central role in legitimising Design Thinking internally, building support among employees, and ensuring continuity beyond isolated projects.

Consultants and facilitators supporting SME innovation should avoid positioning Design Thinking as a rigid methodology and instead present it as a flexible process that supports strategic goals. Effective facilitation in SMEs requires sensitivity to time constraints, multifunctional teams, and varying levels of familiarity with innovation practices. Consultants may also need to invest time in reframing Design Thinking for sceptical audiences and in linking the process to tangible business outcomes.

At the policy level, the findings point to the need for innovation programmes that reflect the realities of SMEs. Frameworks and training materials developed for large firms often fail to resonate with smaller organisations. Public support mechanisms could benefit from modular formats, peer learning, and digital toolkits specifically designed for SME adoption. Web-based Design Thinking resources that are tailored to the specific constraints of SMEs can enhance both accessibility and scalability of design-led innovation practices.

In summary, this study provides practical evidence that Design Thinking can support innovation in SMEs, provided it aligns with internal culture, is supported by leadership, and is embedded in flexible ways that reflect operational realities.

6 Conclusions

This study addressed the research question of how Design Thinking is implemented in small and medium-sized enterprises (SMEs) and which contextual factors shape its outcomes.

The findings identify key enablers for effective Design Thinking implementation, including leadership involvement, iterative experimentation, cross-functional collaboration, and alignment with organisational strategy and culture. Sustainability, while rarely a predefined goal, emerges as an indirect outcome of design-led innovation within SMEs.

The study contributes to the existing literature by providing findings from SME contexts, which remain underrepresented in Design Thinking research. It shows how SMEs translate Design Thinking principles into practice through low-cost prototyping, simplified workshops, and

the engagement of informal innovation champions. These findings reinforce the importance of context-sensitive approaches to design-led innovation in both theory development and practical application.

The study has limitations. The small sample of seven expert participants, while suitable for exploratory qualitative analysis, only allow analytic generalization (Yin, 2018). The reliance on retrospective expert accounts may also introduce recall bias, as the data reflect informed interpretations rather than direct observation.

There are several directions for future research. Longitudinal studies could explore how Design Thinking practices evolve over time and how they contribute to organizational transformation in SMEs. Comparative studies between SMEs, start-ups, and large firms would increase the understanding of contextual influences on Design Thinking adoption. Future studies could also use mixed-method approaches combining qualitative insights with quantitative performance metrics, such as innovation outcomes or customer satisfaction, to measure Design Thinking's impact.

In conclusion, this study reinforces the relevance of Design Thinking as a strategic and operational enabler for SMEs, while arguing for adaptive, practitioner-informed frameworks that reflect the realities of SMEs.

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AI Disclosure Statement

The authors acknowledge the use of ChatGPT for language refinement, grammar and improving clarity of expression. The authors confirm that all intellectual content, interpretation, and conclusions are their own, and that AI use did not compromise the originality, accuracy, or scientific integrity of the work. The authors take full responsibility for the content of the research work.

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8 Appendices

Appendix 1. Implementation of a Sustainable Design Thinking Approach in SMEs.

First-order Codes	Second-order Codes	Summarized
Purpose of Design Thinking implementation	Design Thinking as a mission and vision for empathy and communication Transition from a one-off project to broader adaptation Design Thinking for defining problems and generating ideas Introducing Design Thinking for long-term projects Application of Design Thinking depends on the situation Design Thinking intended for regular agile use	Implementation of a Sustainable Design Thinking Approach
Process of Design Thinking use	Difficulties in grasping and applying Design Thinking due to different backgrounds and comfort zone Learning from initial mistakes in interview techniques for Design Thinking Adoption of Design Thinking through training, workshops, and cultural assessment Source of successful and unsuccessful Design Thinking projects & Ongoing growth and transformation in the practice of Design Thinking Trust-based relationships and teamwork are fundamental in Design Thinking Involvement in both project delivery and training workshops for Design Thinking Comprehensive execution of Design Thinking phases from research to testing	
Reasons for using Design Thinking	Design Thinking helps identify customer needs and align products accordingly Positive results, cost savings, and enjoyable process drive the commitment to Design Thinking Customers' interest and importance for clients led to the team's involvement in Design Thinking Strategic value, collaboration, and enjoyment motivate the use of Design Thinking Support and commitment from those in influential positions are crucial for effective adoption of Design Thinking SMEs' adoption of Design Thinking depends on their focus, goals, and connection to the academic world Demand for digital expertise and strategy led to the recognition of a skills gap and the need for Design Thinking Recruitment of digital talent with design, engineering, and Agile expertise to assist clients across industries	

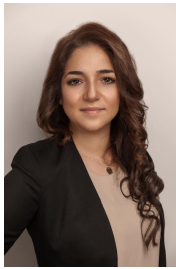
First-order Codes	Second-order Codes	Summarized
Importance of innovation management in SMEs' success	Innovation management needs support from senior management and employee commitment to be effective Change in thinking, especially at the management level, is crucial for successful innovation management Starting with small steps and overcoming budget constraints can be a practical approach for implementing innovation The significance of innovation management depends on the company's size, budget, and industry A clear strategy, values, and leadership are essential for successful innovation management Understanding that innovation is iterative and adopting a Design Thinking mindset are crucial for success Innovation is important for long-term business growth, but its relevance depends on the company's goals & aspirations	
Functional areas in SMEs	Involve employees from various departments, incl. marketing, IT, and management, to bring diverse perspectives and ideas Managers' involvement in Design Thinking workshops can sometimes create complex dynamics that hinder creative work Stakeholders with decision-making power in Design Sprints Diverse cultures and stakeholders lead to better ideas, but it may be more challenging in areas with less cultural diversity All stakeholders related to the problem should be involved, regardless of their role or department Innovation shouldn't be delegated to a specific department & all in the company should be aware of Design Thinking and have a mindset Cross-functional collaboration is crucial for successful innovation management The involvement of different stakeholders and departments depends on the specific goals and focus of the innovation project	
Continual use of Design Thinking in companies	Approach involves working closely with customers, teaching Design Thinking methods, and ensuring they actively engage with the process Design Thinking finds applications in various areas, such as developing new products and features Adoption of Design Thinking varies among organizations, with some using it as a catalyst for new topics or projects There are mixed responses to implementing Design Thinking, with some companies embracing it while others don't continue with it	

First-order Codes	Second-order Codes	Summarized
Continual use of Design Thinking in companies	Projects have evolved into innovation, transformation, agility initiatives & adapting to different organizational cultures Management support plays a crucial role in sustaining Design Thinking practices within companies Iterative processes and research-driven understanding are key elements of applying Design Thinking	
Design Thinking applied to a product	Improved social media and apprentice recruitment for a food manufacturer, reducing turnover Repositioned lab-grown diamonds as an alternative to natural diamonds, emphasizing sustainability and affordability Developed a multi-sensory shower experience as a showcase of digitalization in bathrooms Enhanced branding and customer-centricity for a struggling bank, enabling collaboration and Innovation Created a service for hiring household service providers, iterating based on user feedback Conducted extensive user research, ideated various solutions, and prioritized based on revenue potential	
Failure of innovation	Lack of management involvement from the beginning of a project can hinder success Without Design Thinking, the diamond-themed product may have failed to generate sales Failure to apply Design Thinking could have led to the abandonment of projects in property management services The absence of design research and evaluation could have resulted in the continuation of an unsuitable product Avoiding failure culture and neglecting innovation workshops can impede progress Failure to recognize problems through Design Thinking can lead to ineffective solutions in a robotics company Without user interviews and susceptibility, difficult project decisions may go unaddressed Neglecting Design Thinking intervention could result in the continued development of a poorly performing app Failure to adapt and evolve project goals through Design Thinking can limit problem-solving effectiveness Lack of flexibility and adaptability can hinder the success of the Design Thinking process	

First-order Codes	Second-order Codes	Summarized
Changing goals in Design Thinking	Companies misunderstand Design Thinking as merely brainstorming, failing to grasp its true purpose of understanding people Design Thinking aims to understand people and dispel misconceptions about it being limited to brainstorming While specific areas have improved, the overall goal of Design Thinking remains unchanged Design Thinking has shifted from training to achieving concrete results and solving diverse problems More emphasis is now placed on Design Sprints and aligning problem definition and idea generation The objective is to foster innovation and well-being, but existing structures can hinder progress	
Influence of Design Thinking on Innovation	Design Thinking values people for their unique perspectives, fostering collaboration and personal connections Design Thinking activities are scheduled to allow time for innovation and employees enjoy being creative and making progress Design Thinking sessions can take place within the company or in creative spaces, depending on the context Importance of bottom-up adoption rather than top-down imposition Challenges in instilling a new mindset & starting fresh for successful implementation Addressing doubts and involving skeptics in the process can help them understand and support Design Thinking enables flexibility, agility, and improved processes, which are crucial for SMEs' growth and success Design Thinking enhances employee satisfaction, customer experience, and overall business performance	

Source: Developed by Author

Biographies



Fatma Demir. Fatma Demir is a researcher in business innovation with a focus on design thinking, innovation management, and small and medium-sized enterprises (SMEs). She completed her PhD at the University of Aveiro, where her research examined how design thinking can be systematically structured and integrated into organizational innovation practices. Her work applies mixed-method approaches, including systematic literature reviews, qualitative expert interviews, and Delphi studies. Her research focuses on developing structured, modular frameworks to support the practical implementation of innovation management, aligning with international standards such as ISO 56001. She has published in peer-reviewed journals, including the Journal of Innovation Management, and presented her work at international conferences

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